

Nengneng Yu

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PERSONAL SUMMARY

My research builds **systems for LLMs**: observability and telemetry, fault-tolerant distributed training and serving, and collective communication. A complementary track in **data-driven ML** tackles real-world problems with messy, limited data.

Skills. C, C++, Python, PyTorch, NumPy, Pandas; LLM serving (vLLM, SGLang); distributed systems; GPU and collective communication; machine learning and deep learning.

EDUCATION

University of Maryland, College Park

Ph.D. in Computer Science — GPA: 3.80/4.00

Advisor: Prof. Zaoxing (Alan) Liu

College Park, MD

Aug 2023 – Present

Boston University

B.S. in Computer Engineering, Magna Cum Laude

Boston, MA

Sep 2019 – May 2023

RESEARCH PROJECTS

LLM & Distributed Systems

LLM Internal Observability

Mentor: Prof. Zaoxing (Alan) Liu

Sep 2025 – Present

Froot Lab, UMD

Inference-time workloads increasingly need timely access to a model's internal states, yet today's stacks treat observability as an afterthought; we pursue it as a first-class systems primitive.

- **DMI**: a high-performance, model- and engine-agnostic deep model inspector that decouples internal-state capture from the inference hot path (*on arXiv; poster at NSDI 2026*).

Reliable & Resilient Collective Communication for Distributed ML

Mentor: Prof. Zaoxing (Alan) Liu | Collaborator: Wei Wang

May 2025 – Present

Froot Lab, UMD

Network failures waste 10–15% of GPU hours in large-scale ML clusters, and today's NCCL-style CCLs crash on any error. We close the gap from both the systems and theory sides.

- **R2CC**: an NCCL-compatible CCL that keeps collectives running under inter-node NIC/link failures with negligible overhead (*arXiv preprint; full paper under review*).
- **OptCC**: the first information-theoretic lower bound on AllReduce time under bandwidth-asymmetric topologies, plus a pipelined algorithm that approaches it (*manuscript under review*).

Interactive Research Agents for Internet Incident Investigation

Mentor: Prof. Zaoxing (Alan) Liu

May 2023 – Nov 2023

Froot Lab, UMD

Investigating Internet incidents (outages, BGP misconfigurations, natural-disaster impacts) requires extensive cross-domain expertise and is largely manual today.

- **Generative Research Agent**: an LLM-based agent (Auto-GPT + GPT-4) that simulates an experienced researcher's investigation loop (*HotNets 2023; co-first author*).

Data-Driven ML

AI for Science — Cross-Cohort Biomedical Analysis

Aug 2024 – Jun 2025

Mentors: Prof. Zaoxing (Alan) Liu, Yuefan Wang

Froot Lab, UMD & Johns Hopkins Med.

Biomedical multi-omics is small-sample, high-dimensional, and batch-effect heavy across cohorts; we attack this with generative augmentation and integrative analysis.

- **TabSyM**: a generative pipeline (tabular diffusion + task-aware sampling + multi-domain adversarial alignment) that substantially boosts cancer-prognosis prediction across cohorts (*bioRxiv preprint*; *first author*).
- **15-layer multi-omics gastric cancer dissection**: a multi-institution study identifying therapeutically actionable gastric-cancer ecotypes (*Cell Reports Medicine*; *primary author*).

APT Detection under Concept Drift

Feb 2022 – May 2023

Mentors: Prof. Zaoxing (Alan) Liu, Prof. Tuo Zhao

Red Hat & Boston Univ. & Georgia Tech

ML-based provenance intrusion detection systems suffer concept drift as Advanced Persistent Threats evolve, while post-drift labels are scarce and naive retraining causes catastrophic forgetting.

- **Tidal**: a multi-head Transformer with a pre-train/fine-tune workflow that adapts to new attacks while retaining prior-attack accuracy (*NINeS 2026*).

PUBLICATIONS & PREPRINTS

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- [1] **Nengneng Yu***, Sixian Xiong*, Yibo Zhao, Wei Wang, Zaoxing Liu. “Enabling Performant and Flexible Model-Internal Observability for LLM Inference.” *arXiv preprint*, 2026.
 - [2] **Nengneng Yu**, Sixian Xiong, Yibo Zhao, Wei Wang, Zaoxing Liu. “DMI: Performant and Flexible Deep Model Inspector for LLM Inference.” *Poster, USENIX NSDI 2026*.
 - [3] Wei Wang, **Nengneng Yu**, Sixian Xiong, Zaoxing Liu. “Reliable and Resilient Collective Communication Library for LLM Training and Serving.” *arXiv preprint*, 2025.
 - [4] **Nengneng Yu**, Yuefan Wang, Lindsey Kathleen Olsen, Bing Zhang, Hui Zhang, Zaoxing Liu. “TabSyM: A Generative Pipeline for Small Multi-Cohort Omics Tabular Data.” *bioRxiv preprint*, 2025.
 - [5] Yuefan Wang*, Lindsey Kathleen Olsen*, ..., **Nengneng Yu**, ..., Bing Zhang. “A 15-Layer Multi-Omics Dissection of Gastric Cancer Ecotypes Reveals Therapeutic Opportunities.” *Cell Reports Medicine*, 2026. (**Nengneng Yu** as primary author.)
 - [6] Yajie Zhou*, **Nengneng Yu***, Zaoxing Liu. “Towards Interactive Research Agents for Internet Incident Investigation.” *HotNets 2023*. (* co-first authors.)
 - [7] Yajie Zhou, **Nengneng Yu**, Tuo Zhao, Zaoxing Liu. “Tidal: Tackling Concept Drift in Provenance-Based Advanced Persistent Threats Detection.” *NINeS 2026*.